

3ai	B1: correct from $t=0\text{s}$ to $t=6.0\text{s}$ B1: correct from $t=6.0\text{s}$ to $t=9.0\text{s}$	B1 B1
3aii	From Fig 3.2, $t = 3.0\text{ s}$ to $t = 4.2\text{ s}$, the increase in gravitational potential energy is $\Delta GPE = 130 - 60 = 70\text{ kJ}$ $mg\Delta h = 70\,000$ $\Delta h = \frac{70\,000}{380 \times 9.81}$ $= 18.78 \approx 19\text{ m}$ The increase in height, and therefore the diameter d of loop, is 19 m.	C1 A1
3aiii	$KE \text{ at top of loop} = 310 - 130 = 180\text{ kJ}$ $\frac{1}{2}mv^2 = 180\,000$ $mv^2 = 2(180\,000)$ $F_c = \frac{mv^2}{r}$ $= \frac{2(180\,000)}{\frac{18.78}{2}}$	C1 C1

	$N + mg = 38\,339$ $N = 38339 - (380)(9.81)$ $N = 34611 \approx 35\,000\text{ N}$	A1
3bi	$\text{rate of loss of KE} = \frac{\text{loss of KE}}{\text{duration in zone A}}$ $= \frac{(310 - 250) \times 10^3}{7.4 - 6.0}$ $= 42857 \approx 43\,000\text{ W}$	C1 A1
3bii	$P = Fv$ $F = \frac{42857}{34}$ $= 1260\text{ N}$	C1 A1